MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL, 4-BIT BINARY FULL ADDERS WITH FAST CARRY, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, 4-bit binary full adders with fast carry. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-bit binary full adders with fast carry
02	4-bit binary full adders with fast carry

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8", dual-in-line package)
F	F-5 (16-lead, 1/4" x 3/8", flat package)
X	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA dc to +5.5 V dc
Storage temperature range	-65° to +150°C
Maximum power dissipation (P_D) 1/	214 mW dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC}):	
Case E	50°C/W
Case F	70°C/W
Case X	60°C/W
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.7 V dc

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Normalized fanout (each output):	
Logical low level- - - - - - - - - - - - - - -	10 maximum
Logical high level - - - - - - - - - - - - - - -	20 maximum
Case operating temperature range (T_C)- - - -	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be in accordance with MIL-M-38510 and as specified herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth table. The truth table shall be as specified on figure 2.

3.2.3 Logic diagrams. The logic diagrams shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III. Subgroups 7 and 8 testing requires only a summary of attributes data.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Unit
			Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V; I_{OH} = -400 µA	2.5	---	V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.7 V, V_{IH} = 2.0 V, I_{OL} = 4 mA	---	0.4	V
Input diode clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA, T_C = +25°C	---	-1.5	V
Low-level input current at any A or B input	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-0.24	-0.8	mA
Low-level input current at C_0 input	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-0.12	-0.40	mA
High-level input current at any A or B input	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	---	40	µA
High-level input current at C_0 input	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	---	20	µA
High-level input current at any A or B input	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	---	200	µA
High-level input current at C_0 input	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	---	100	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V, all inputs = 5.5 V 1/	-15	-100	mA
Supply current	I_{CC1}	All inputs = Ground, V_{CC} = 5.5 V	---	39	mA
	I_{CC2}	All B inputs = Ground, all other inputs = 5.5 V, V_{CC} = 5.5 V	---	34	mA
	I_{CC3}	All inputs = 5.5 V, V_{CC} = 5.5 V	---	34	mA
Propagation delay time, high-to-low level, C_0 to Σ_1	t_{PHL1}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, low-to-high level, C_0 to Σ_1	t_{PLH1}	C_L = 50 pF, R_L = 2 kΩ	5	39	ns
Propagation delay time, high-to-low level, C_0 to Σ_2	t_{PHL2}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, low-to-high level, C_0 to Σ_2	t_{PLH2}	C_L = 50 pF, R_L = 2 kΩ	5	39	ns

See footnote at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Unit
			Min	Max	
Propagation delay time, high-to-low level, C_0 to Σ_3	t_{PHL3}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, low-to-high level, C_0 to Σ_3	t_{PLH3}	C_L = 50 pF, R_L = 2 kΩ	5	39	ns
Propagation delay time, high-to-low level, C_0 to Σ_4	t_{PHL4}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, low-to-high level, C_0 to Σ_4	t_{PLH4}	C_L = 50 pF, R_L = 2 kΩ	5	39	ns
Propagation delay time, high-to-low level, C_0 to C_4	t_{PHL5}	C_L = 50 pF, R_L = 560Ω	5	35	ns
Propagation delay time, low-to-high level, C_0 to C_4	t_{PLH5}	C_L = 50 pF, R_L = 560Ω	5	32	ns
Propagation delay time, high-to-low level, A_2 to Σ_2	t_{PHL6}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, low-to-high level, A_2 to Σ_2	t_{PLH6}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, high-to-low level, B_2 to Σ_2	t_{PHL7}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, low-to-high level, B_2 to Σ_2	t_{PLH7}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, high-to-low level, A_4 to Σ_4	t_{PHL8}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, low-to-high level, A_4 to Σ_4	t_{PLH8}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, high-to-low level, B_4 to Σ_4	t_{PHL9}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns
Propagation delay time, low-to-high level, B_4 to Σ_4	t_{PLH9}	C_L = 50 pF, R_L = 2 kΩ	5	40	ns

1/ Not more than one output should be shorted at a time.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1, 2, 3
Additional electrical subgroups for group C periodic inspection	N/A	N/A
Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1 (see 4.2c).

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-35810, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

b. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

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c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

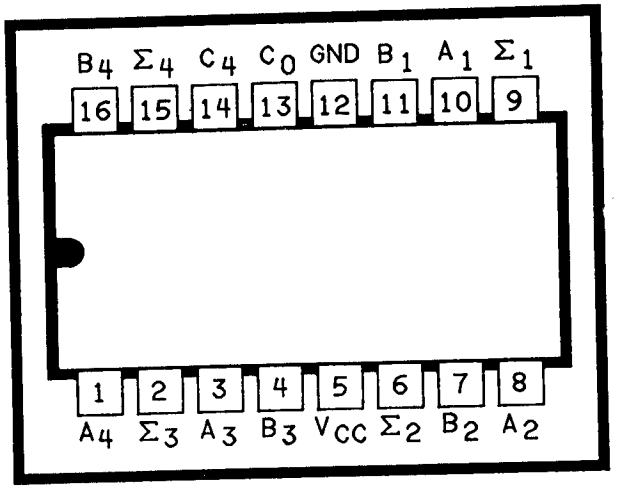
4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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Device type 01

Cases E and F



Device type 01

Case X

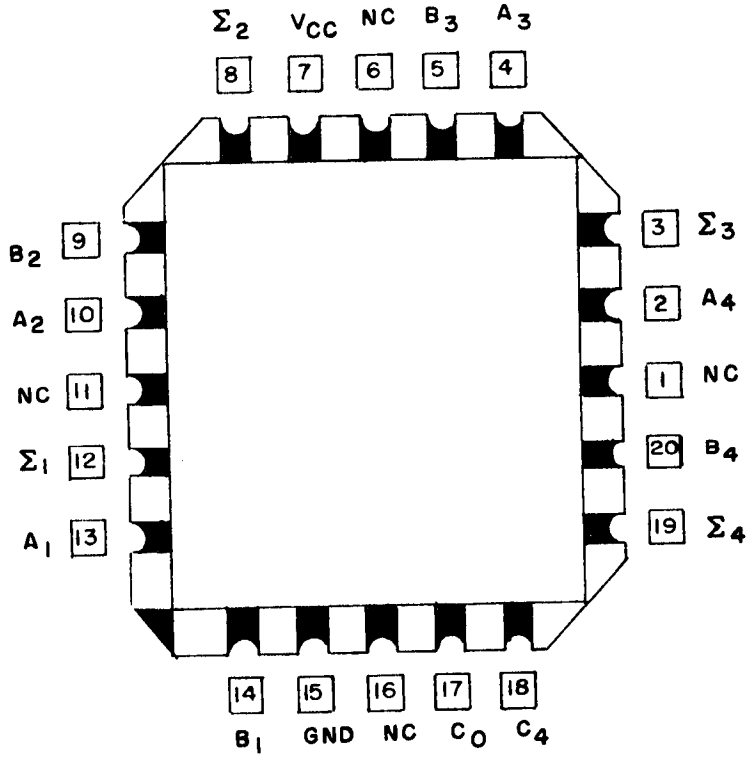
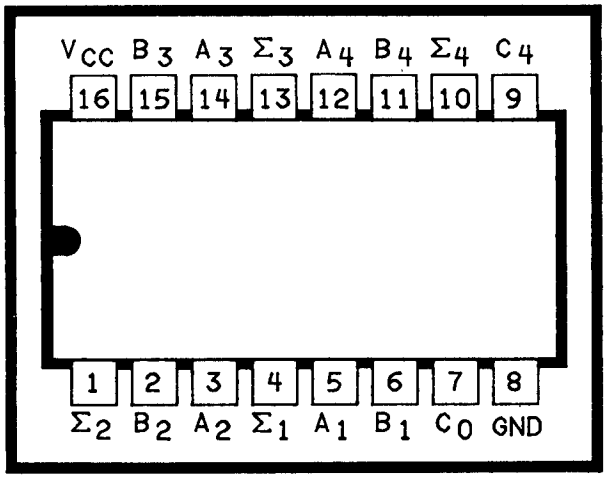


FIGURE 1. Terminal connections (top views).

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Device type 02

Cases E and F



Device type 02

Case X

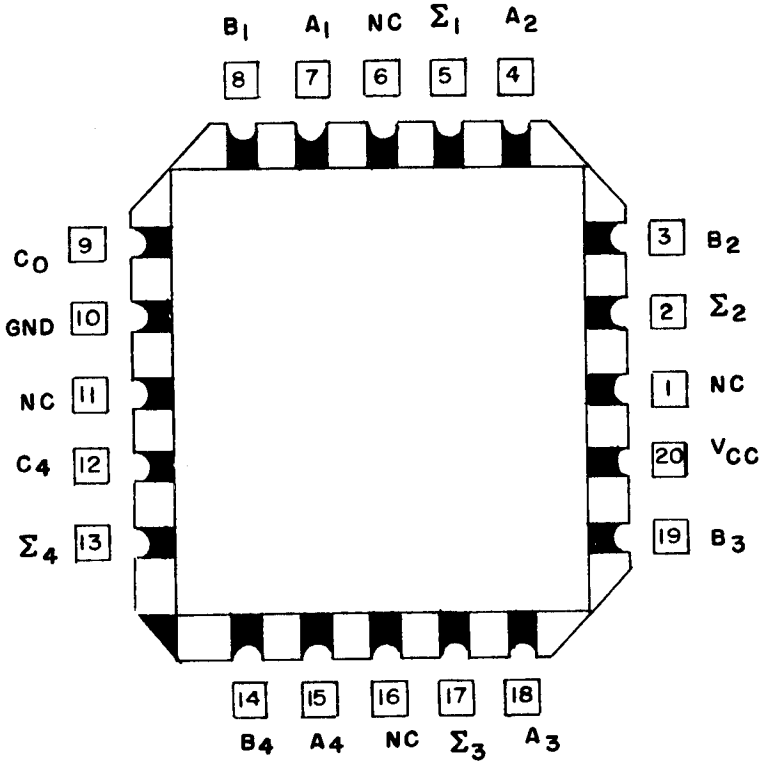


FIGURE 1. Terminal connections (top views) - Continued.

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Device types 01 and 02

INPUT				OUTPUT					
				When C_0 = L			When C_0 = H		
				When C_2 = L			When C_2 = H		
A_1 / A_3	B_1 / B_3	A_2 / A_4	B_2 / B_4	Σ_1 / Σ_3	Σ_2 / Σ_4	C_2 / C_4	Σ_1 / Σ_3	Σ_2 / Σ_4	C_2 / C_4
L	L	L	L	L	L	L	H	L	L
H	L	L	L	H	L	L	L	H	L
L	H	L	L	H	L	L	L	H	L
H	H	L	L	L	H	L	H	H	L
L	L	H	L	L	H	L	H	H	L
H	L	H	L	H	H	L	L	L	H
L	H	H	L	H	H	L	L	L	H
H	H	H	L	L	L	H	H	L	H
L	L	L	H	L	H	L	H	H	L
H	L	L	H	H	H	L	L	L	H
L	H	L	H	H	H	L	L	L	H
H	H	L	H	L	L	H	H	L	H
L	L	H	H	L	L	H	H	L	H
H	L	H	H	H	L	H	L	H	H
L	H	H	H	H	L	H	L	H	H
H	H	H	H	L	H	H	H	H	H

NOTE: Input conditions at A_1, B_1, A_2, B_2, and C_0 are used to determine outputs Σ_1 and Σ_2 and the value of the internal carry C_2. The values at C_2, A_3, B_3, A_4, and B_4 are then used to determine outputs Σ_3, Σ_4, and C_4.

FIGURE 2. Truth table.

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Device types 01 and 02

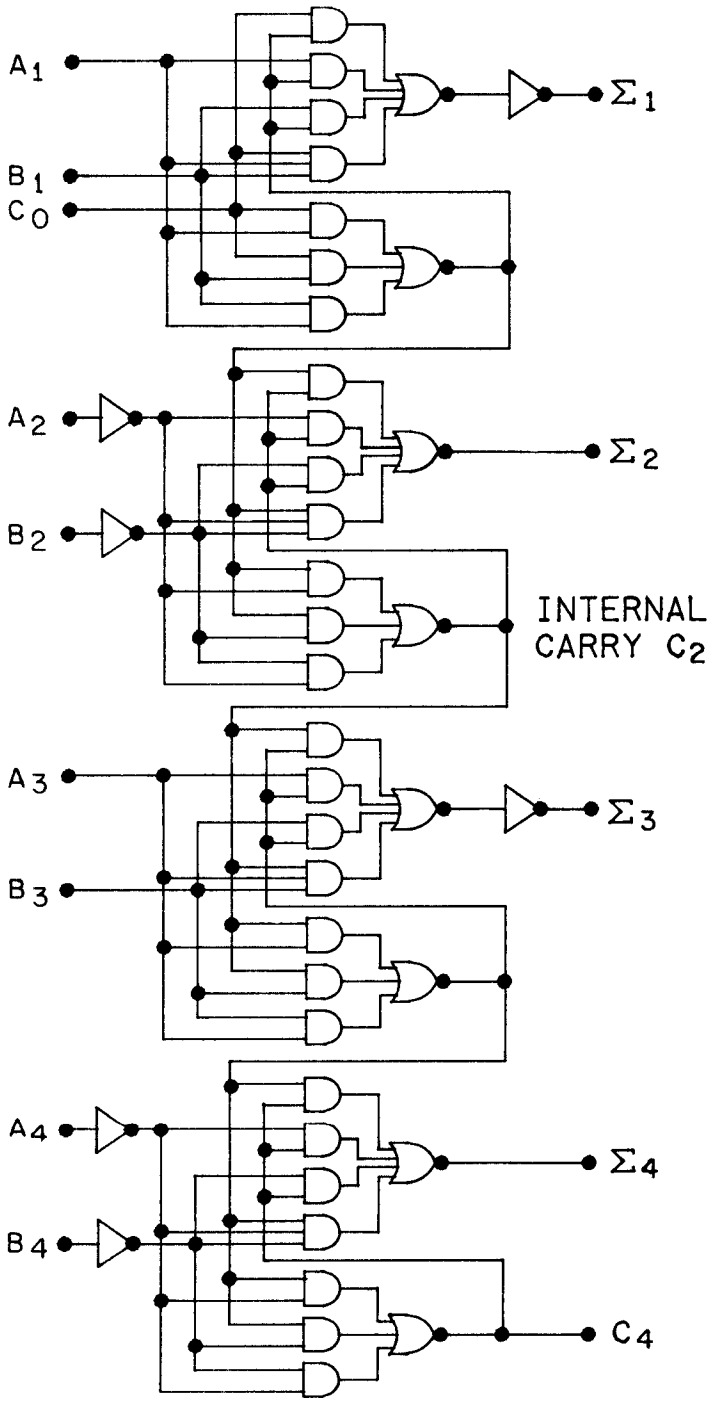


FIGURE 3. Logic diagrams.

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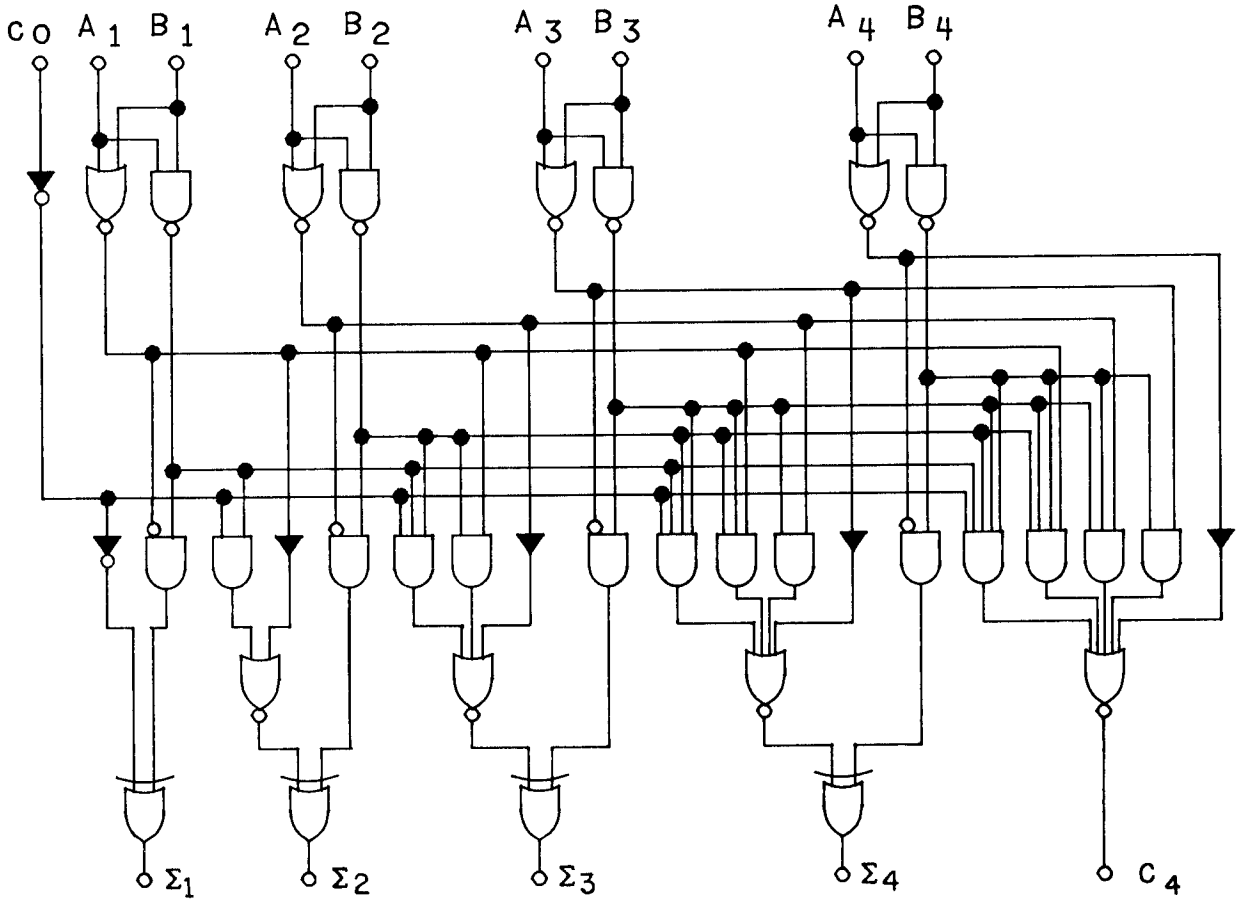
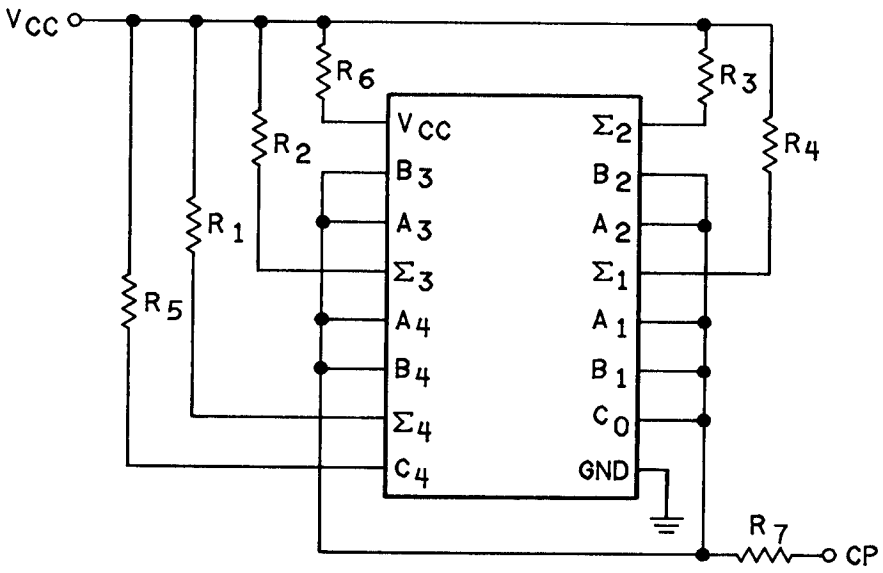


FIGURE 3. Logic diagrams- continued.

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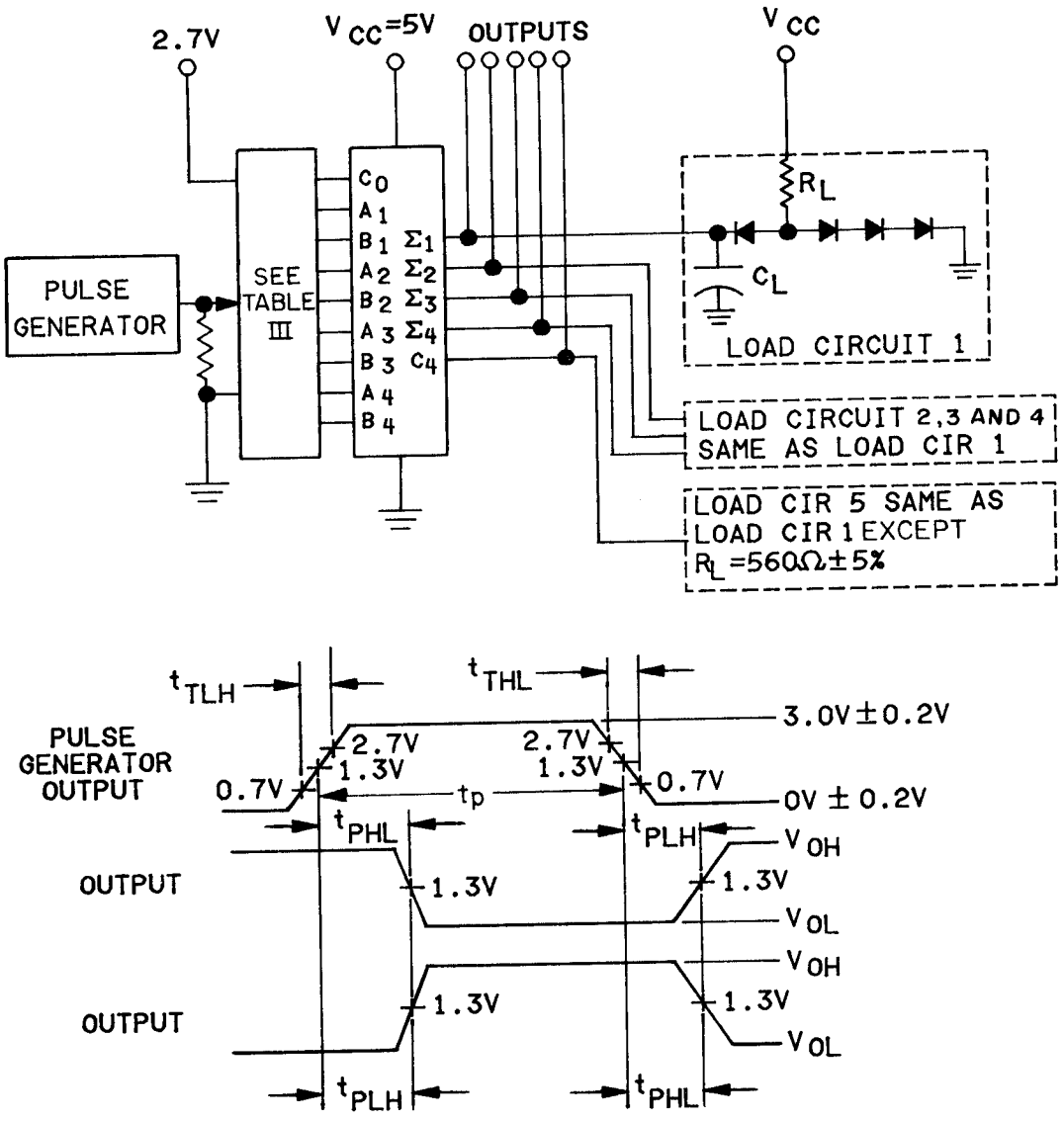
Device Type 01 and 02



NOTES:
1. CP = 100 kHz ±10%; 50% duty cycle; V_{IH} = 2.0 V minimum to 5.5 V maximum; V_{IL} = -0.5 V minimum to 0.7 V maximum.
2. R1 through R5 = 1 kΩ ±5%, R6 = 10Ω ±5% through R7 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure 5.0 V minimum at device V_{CC} terminal.

FIGURE 4. Burn-in and steady state life test circuit.

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NOTES:
1. The pulse generator has the following characteristics: PRR $\leq$ 1 MHz, $t_{TLH} \leq$ 15 ns, $t_{THL} \leq$ 6 ns, t_p = 200 ns and $Z_{OUT} \cong$ 50Ω.
2. C_L = 50 pF ±10%, including scope probe, wiring, and stray capacitance without package in test fixture.
3. All diodes are 1N3064 or equivalent.
4. R_L = 2 kΩ ±5%.
5. Load circuits on a given output are only required when specified by "OUT" on the specific test in table III. Load circuits may otherwise be omitted.

FIGURE 5. Switching time test circuit for device types 01 and 02.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	A_4	Σ_3	A_3	B_3	V_{CC}	Σ_2	B_2	A_2	Σ_1	A_1	B_1	GND	C_0	C_4	Σ_4	B_4		Min	Max	
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5	2.0 V " " " "	-.4 mA	2.0 V " " " "	2.0 V " " " "	4.5 V " " " "	-.4 mA	2.0 V " " " "	2.0 V " " " "	-.4 mA	2.0 V " " " "	2.0 V " " " "	GND " " " "	2.0 V " " " "	-.4 mA	-.4 mA	2.0 V " " " "	Σ_1 Σ_2 Σ_3 Σ_4 C_4	2.5 " " " "		V " " " "
	V_{OL}	3007	6 7 8 9 10	0.7 V " " " "	4 mA	0.7 V " " " "	0.7 V " " " "	" " " " "	4 mA	0.7 V " " " "	0.7 V " " " "	4 mA	0.7 V " " " "	0.7 V " " " "	" " " " "	0.7 V " " " "	4 mA	4 mA	0.7 V " " " "	Σ_1 Σ_2 Σ_3 Σ_4 C_4		0.4 " " " "	" " " " "
	V_{IC}		11 12 13 14 15 16 17 18 19	-18 mA		-18 mA	-18 mA	" " " " " " " " "		-18 mA	-18 mA		-18 mA	-18 mA	" " " " " " " " "	-18 mA			-18 mA	A_4 A_3 B_3 B_2 A_2 A_1 B_1 C_0 B_4		-1.5 " " " " " " " "	" " " " " " " " "
	I_{IH1}	3010	20 21 22 23 24 25 26 27	2.7 V GND		2.7 V GND	GND 2.7 V	5.5 V " " " " " " "		2.7 V GND	GND 2.7 V		2.7 V GND	GND 2.7 V	" " " " " " " "				GND 2.7 V	A_3 B_3 A_1 B_1 A_4 B_2 A_2 B_4		40 " " " " " " "	µA " " " " " " "
	I_{IH2}	"	28					"							"	2.7 V				C_0		20	"
	I_{IH3}	" " " " " " " "	29 30 31 32 33 34 35 36	5.5 V GND		5.5 V GND	GND 5.5 V	" " " " " " " "		5.5 V GND	GND 5.5 V		5.5 V GND	GND 5.5 V	" " " " " " " "				GND 5.5 V	A_3 B_3 A_1 B_1 A_4 B_2 A_2 B_4		200 " " " " " " "	" " " " " " " "
	I_{IH4}	"	37					"							"	5.5 V				C_0		100	"
	I_{IL1}	3009 " " " " " " "	38 39 40 41 42 43 44 45	0.4 V 5.5 V		0.4 V 5.5 V	5.5 V 0.4 V	" " " " " " " "		0.4 V 5.5 V	5.5 V 0.4 V		0.4 V 5.5 V	5.5 V 0.4 V	" " " " " " " "				5.5 V 0.4 V	A_3 B_3 A_1 B_1 A_4 B_2 A_2 B_4	2/ " " " " " " "	2/ " " " " " " "	mA " " " " " " "
	I_{IL2}	"	46					"							"	0.4 V				C_0	"	"	"
	I_{OS} 3/	3011 " " " "	47 48 49 50 51	5.5 V " " " "	GND	5.5 V " " " "	5.5 V " " " "	" " " " "	GND	5.5 V " " " "	5.5 V " " " "	GND	5.5 V " " " "	5.5 V " " " "	" " " " "	5.5 V " " " "	GND	GND	5.5 V " " " "	Σ_3 Σ_2 Σ_1 Σ_4 C_4	-15 " " " "	-100 " " " "	" " " " "

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
(Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20	Measured terminal	Test limits		Unit
			Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Test no.	A_4	Σ_3	A_3	B_3	V_{CC}	Σ_2	B_2	A_2	Σ_1	A_1	B_1	GND	C_0	C_4	Σ_4	B_4		Min	Max	
1 T_C = +25°C	I_{CC1}	3005	52	GND		GND	GND	5.5 V		GND	GND		GND	GND	"	GND			GND	V_{CC}		39	mA
	I_{CC2}	"	53	5.5 V		5.5 V	GND	"		GND	5.5 V		5.5 V	GND	"	5.5 V			GND	"		34	"
	I_{CC3}	"	54	5.5 V		5.5 V	5.5 V	"		5.5 V	5.5 V		5.5 V	5.5 V	"	5.5 V			5.5 V	"		34	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C 4/	Truth table tests	3014	55	B	L	B	B	5.0 V	L	B	B	L	B	B	GND	B	L	L	B	All outputs	See 5/ and 6/		
		"	56	A	L	A	A	"	L	"	"	H	B	A	"	A	H	L	A	"	"		
		"	57	"	H	B	B	"	L	"	"	L	A	"	"	B	L	H	"	"	"		
		"	58	"	H	A	B	"	H	"	"	L	"	"	"	"	H	H	"	"	"		
		"	59	"	L	B	A	"	L	"	"	L	"	"	"	"	L	H	"	"	"		
		"	60	B	L	A	A	"	H	"	"	L	"	"	"	"	L	"	"	"	"		
		"	61	"	H	B	A	"	"	"	"	L	"	"	"	"	H	"	"	"	"		
		"	62	"	H	A	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	63	"	L	B	A	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	64	A	L	A	A	"	"	"	A	L	"	"	"	"	H	"	"	"	"		
		"	65	"	H	B	A	"	"	"	"	L	"	"	"	"	H	"	"	"	"		
		"	66	"	H	A	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	67	"	H	B	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	68	B	L	B	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	69	B	H	A	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	70	A	H	B	A	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	71	A	L	A	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	72	B	L	B	A	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	73	B	H	A	A	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	74	B	L	A	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	75	A	H	B	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	76	A	L	B	A	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	77	A	H	B	A	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	78	B	H	A	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	79	"	L	B	A	"	"	"	A	L	"	"	"	"	L	"	"	"	"		
		"	80	"	L	A	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	81	"	H	B	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	82	A	H	A	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	83	A	L	B	A	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	84	A	L	A	B	"	"	"	"	L	"	"	"	"	L	"	"	"	"		
		"	85	B	L	B	B	"	H	"	"	H	B	A	"	"	H	H	B	"	"		
		"	86	B	H	B	B	"	H	"	"	H	A	"	"	"	L	L	B	"	"		
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	87					5.0 V		GND	GND	OUT	2.7 V	GND	GND	IN				C_0 to Σ_1	5	30	ns
	t_{PLH2}		88					"	OUT	"	2.7 V		"	"	"	"				C_0 to Σ_2	"	"	"
	t_{PLH3}	"	89		OUT	2.7 V	GND	"		"	"		"	"	"	"				C_0 to Σ_3	"	"	"
	t_{PLH4}	"	90	2.7 V		2.7 V	GND	"		"	"		"	"	"	"	OUT	OUT	GND	C_0 to Σ_4	"	"	"
	t_{PLH5}	"	91	2.7 V		2.7 V	GND	"		"	"		"	"	"	"			GND	C_0 to C_4	"	25	"
	t_{PLH6}	"	92					"	OUT	"	IN		GND	"	"	GND				A_2 to Σ_2	"	35	"
	t_{PLH7}	"	93					"	OUT	IN	GND		GND	"	"	GND		OUT	GND	B_2 tp Σ_2	"	"	"
	t_{PLH8}	"	94	IN		GND	GND	"		"	"		"	"	"	"		OUT	GND	A_4 to Σ_4	"	"	"
	t_{PLH9}	"	95	GND		GND	GND	"		"	"		"	"	"	"		OUT	IN	B_4 to Σ_4	"	"	"
	t_{PHL1}	"	96					"		GND	GND	OUT	2.7 V	GND	"	IN				C_0 to Σ_1	"	"	"
	t_{PHL2}	"	97					"	OUT	"	2.7 V		"	"	"	"				C_0 to Σ_2	"	"	"
	t_{PHL3}	"	98		OUT	2.7 V	GND	"		"	"		"	"	"	"				C_0 to Σ_3	"	"	"
	t_{PHL4}	"	99	2.7 V		2.7 V	GND	"		"	"		"	"	"	"	OUT	OUT	GND	C_0 to Σ_4	"	"	"
	t_{PHL5}	"	100	2.7 V		2.7 V	GND	"		"	"		"	"	"	"			GND	C_0 to C_4	"	28	"
	t_{PHL6}	"	101					"	OUT	"	IN		GND	"	"	GND				A_2 to Σ_2	"	35	"

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case X 1/ Cases E,F Test no.	1 A_4	2 Σ_3	3 A_3	4 B_3	5 V_{CC}	6 Σ_2	7 B_2	8 A_2	9 Σ_1	10 A_1	11 B_1	12 GND	13 C_0	14 C_4	15 Σ_4	16 B_4	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHL7}	3003	102					5.0 V	OUT	IN	GND		GND	GND	GND	GND				B_2 to Σ_2	5	35	ns
	t_{PHL8}	Fig. 5	103	IN		GND	GND	"							"			OUT	GND	A_4 to Σ_4	"	35	"
	t_{PHL9}		104	GND		GND	GND	"							"			OUT	IN	B_4 to Σ_4	"	35	"
10 T_C = +125°C	t_{PLH1}	"	105					"		GND	GND	OUT	2.7 V	GND	"	IN				C_0 to Σ_1	"	39	"
	t_{PLH2}	"	106					"	OUT	"	2.7 V		"	"	"	"				C_0 to Σ_2	"	"	"
	t_{PLH3}	"	107		OUT	2.7 V	GND	"		"	"		"	"	"	"				C_0 to Σ_3	"	"	"
	t_{PLH4}	"	108	2.7 V		2.7 V	GND	"		"	"		"	"	"	"		OUT	GND	C_0 to Σ_4	"	"	"
	t_{PLH5}	"	109	2.7 V		2.7 V	GND	"		"	"		"	"	"	"	OUT		GND	C_0 to C_4	"	32	"
	t_{PLH6}	"	110					"	OUT	"	IN		GND	"	"	GND				A_2 to Σ_2	"	40	"
	t_{PLH7}	"	111					"	OUT	IN	GND		GND	"	"	GND				B_2 tp Σ_2	"	"	"
	t_{PLH8}	"	112	IN		GND	GND	"							"			OUT	GND	A_4 to Σ_4	"	"	"
	t_{PLH9}	"	113	GND		GND	GND	"							"			OUT	IN	B_4 to Σ_4	"	"	"
	t_{PHL1}	"	114					"		GND	GND	OUT	2.7 V	GND	"	IN				C_0 to Σ_1	"	"	"
	t_{PHL2}	"	115					"	OUT	"	2.7 V		"	"	"	"				C_0 to Σ_2	"	"	"
	t_{PHL3}	"	116		OUT	2.7 V	GND	"		"	"		"	"	"	"				C_0 to Σ_3	"	"	"
	t_{PHL4}	"	117	2.7 V		2.7 V	GND	"		"	"		"	"	"	"		OUT	GND	C_0 to Σ_4	"	"	"
	t_{PHL5}	"	118	2.7 V		2.7 V	GND	"		"	"		"	"	"	"	OUT		GND	C_0 to C_4	"	35	"
	t_{PHL6}	"	119					"	OUT	"	IN		GND	"	"	GND				A_2 to Σ_2	"	40	"
	t_{PHL7}	"	120					"	OUT	IN	GND		GND	"	"	GND				B_2 to Σ_2	"	"	"
	t_{PHL8}	"	121	IN		GND	GND	"							"			OUT	GND	A_4 to Σ_4	"	"	"
	t_{PHL9}	"	122	GND		GND	GND	"							"			OUT	IN	B_4 to Σ_4	"	"	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Case X pin not referenced are NC.

2/ Test limits are as follows:

Symbol	Min/Max limits (mA) for circuit: A	B	C	D	E	F
I_{IL1}	-.32/-.80	-.24/-.72	-.33/-.71	-.32/-.80	-.24/-.72	-.30/-.76
I_{IL2}	-.16/-.40	-.12/-.36	-.15/-.38	-.16/-.40	-.12/-.36	-.15/-.38

3/ Alternate I_{OS} test conditions for I_{OS} test shall be as follows: Tests 47, 48, 49, 50, 51; pins Σ_3, Σ_2, Σ_1, Σ_4, C_4 respectively, shall be forced to 0.5 V. Tests 47 through 51, V_{CC} pin shall be 6.0 V.

4/ Test numbers 55 through 86 shall be run in sequence.

5/ A = ≥ 2.0 V; B = ≤0.7 V.

6/ Output voltages shall be either:
a. H ≥2.5 V minimum and L ≤0.4 V maximum when using a high speed checker double comparator.
b. H ≥1.5 V minimum and L ≤1.5 V maximum when using a high speed checker single comparator.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case X 1/ / Cases E,F / Test no.	1 / Σ_2	2 / B_2	3 / A_2	4 / Σ_1	5 / A_1	6 / B_1	7 / C_0	8 / GND	9 / C_4	10 / Σ_4	11 / B_4	12 / A_4	13 / Σ_3	14 / A_3	15 / B_3	16 / V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	-.4 mA	2.0 V	2.0 V	-.4 mA	2.0 V	2.0 V	2.0 V	GND	-.4 mA	-.4 mA	2.0 V	2.0 V	-.4 mA	2.0 V	2.0 V	4.5 V	Σ_1 Σ_2 Σ_3 Σ_4 C_4	2.5		V
		"	2		"	"		"	"	"	"			"	"		"	"	"		"		
		"	3		"	"		"	"	"	"			"	"		"	"	"		"		
		"	4		"	"		"	"	"	"			"	"		"	"	"		"		
		"	5		"	"		"	"	"	"			"	"		"	"	"		"		
	V_{OL}	3007	6	4 mA	0.7 V	0.7 V	4 mA	0.7 V	0.7 V	0.7 V	"	4 mA	4 mA	0.7 V	0.7 V	4 mA	0.7 V	0.7 V	"	Σ_1 Σ_2 Σ_3 Σ_4 C_4		0.4	"
		"	7		"	"		"	"	"	"			"	"		"	"	"		"		
		"	8		"	"		"	"	"	"			"	"		"	"	"		"		
		"	9		"	"		"	"	"	"			"	"		"	"	"		"		
		"	10		"	"		"	"	"	"			"	"		"	"	"		"		
	V_{IC}		11		-18 mA	-18 mA		-18 mA	-18 mA	-18 mA	"			-18 mA	-18 mA		-18 mA	-18 mA	"	B_2 A_2 A_1 B_1 C_0 B_4 A_4 A_3 B_3		-1.5	"
			12								"								"		"		
			13								"								"		"		
			14								"								"		"		
			15								"								"		"		
			16								"								"		"		
			17								"								"		"		
			18								"								"		"		
			19								"								"		"		
	I_{IH1}	3010	20		GND	2.7 V		2.7 V	GND		"			GND	2.7 V		2.7 V	GND	5.5 V	A_2 B_2 A_1 B_1 A_4 B_4 A_3 B_3		40	μA
		"	21		2.7 V	GND		GND	2.7 V		"			2.7 V	GND		GND	2.7 V	"			"	
		"	22		"	"		"	"		"			"	"		"	"	"		"		
		"	23		"	"		"	"		"			"	"		"	"	"		"		
		"	24		"	"		"	"		"			"	"		"	"	"		"		
		"	25		"	"		"	"		"			"	"		"	"	"		"		
		"	26		"	"		"	"		"			"	"		"	"	"		"		
		"	27		"	"		"	"		"			"	"		"	"	"		"		
	I_{IH2}	"	28							2.7 V	"								"	C_0		20	"
	I_{IH3}	"	29		GND	5.5 V		5.5 V	GND		"			GND	5.5 V		5.5 V	GND	"	A_2 B_2 A_1 B_1 A_4 B_4 A_3 B_3		200	"
		"	30		"	"		"	"		"			"	"		"	"	"		"		
		"	31		"	"		"	"		"			"	"		"	"	"		"		
		"	32		"	"		"	"		"			"	"		"	"	"		"		
		"	33		"	"		"	"		"			"	"		"	"	"		"		
		"	34		"	"		"	"		"			"	"		"	"	"		"		
		"	35		"	"		"	"		"			"	"		"	"	"		"		
		"	36		"	"		"	"		"			"	"		"	"	"		"		
	I_{IH4}	"	37							5.5 V	"								"	C_0		100	"
	I_{IL1}	3009	38		5.5 V	0.4 V		0.4 V	5.5 V		"			5.5 V	0.4 V		0.4 V	5.5 V	"	A_2 B_2 A_1 B_1 A_4 B_4 A_3 B_3	2/	2/	mA
		"	39		0.4 V	5.5 V		5.5 V	0.4 V		"			0.4 V	5.5 V		5.5 V	0.4 V	"		"	"	"
		"	40		"	"		"	"		"			"	"		"	"	"		"	"	"
		"	41		"	"		"	"		"			"	"		"	"	"		"	"	"
		"	42		"	"		"	"		"			"	"		"	"	"		"	"	"
		"	43		"	"		"	"		"			"	"		"	"	"		"	"	"
		"	44		"	"		"	"		"			"	"		"	"	"		"	"	"
		"	45		"	"		"	"		"			"	"		"	"	"		"	"	"
	I_{IL2}	"	46							0.4 V	"								"	C_0	"	"	"
	I_{OS} 3/	3011	47	GND	5.5 V	5.5 V	GND	5.5 V	5.5 V	5.5 V	"	GND	GND	5.5 V	5.5 V	GND	5.5 V	5.5 V	"	Σ_2 Σ_1 Σ_4 Σ_3 C_4	-15	-100	"
		"	48		"	"		"	"	"	"			"	"		"	"	"		"	"	
		"	49		"	"		"	"	"	"			"	"		"	"	"		"	"	
		"	50		"	"		"	"	"	"			"	"		"	"	"		"	"	
		"	51		"	"		"	"	"	"			"	"		"	"	"		"	"	

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case X 1/ Cases E,F Test no.	1 Σ2	2 B2	3 A2	4 Σ1	5 A1	6 B1	7 C0	8 GND	9 C4	10 Σ4	11 B4	12 A4	13 Σ3	14 A3	15 B3	16 VCC	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	I_{CC1}	3005	52		GND	GND		GND	GND	GND	GND			GND	GND		GND	GND	5.5 V	V_{CC}		39	mA
	I_{CC2}	"	53		GND	5.5 V		5.5 V	GND	5.5 V	"			GND	5.5 V		5.5 V	GND	"	"		34	"
	I_{CC3}	"	54		5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	"			5.5 V	5.5 V		5.5 V	5.5 V	"	"		34	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C 4/	Truth table test	3014	55	L	B	B	L	B	B	B	GND	L	L	B	B	L	B	B	5.0 V	All outputs	See 5/ and 6/		
		"	56	L	"	"	H	B	"	A	"	H	H	A	A	L	A	A	"	"			
		"	57	L	"	"	H	A	"	B	"	"	L	"	"	H	B	A	"	"			
		"	58	H	"	"	L	A	"	A	"	"	"	"	"	H	A	B	"	"			
		"	59	L	"	"	H	B	A	B	"	"	"	"	"	L	B	B	"	"			
		"	60	H	"	"	L	B	"	A	"	"	"	"	B	L	A	A	"	"			
		"	61	"	"	"	L	A	"	B	"	L	H	"	"	H	B	A	"	"			
		"	62	"	"	"	H	A	"	A	"	L	H	"	"	H	A	B	"	"			
		"	63	"	"	A	L	B	B	B	"	L	L	"	"	L	B	B	"	"			
		"	64	"	"	"	H	B	"	A	"	H	L	B	A	L	A	A	"	"			
		"	65	"	"	"	H	A	"	B	"	L	H	B	"	H	B	A	"	"			
		"	66	L	"	"	L	A	"	A	"	H	L	A	"	L	A	A	"	"			
		"	67	H	"	"	H	B	A	B	"	L	L	B	"	L	B	A	"	"			
		"	68	L	"	"	L	B	"	A	"	"	"	"	B	H	B	B	"	"			
		"	69	L	"	"	L	A	"	B	"	"	"	"	B	H	A	B	"	"			
		"	70	L	"	"	H	A	"	A	"	"	"	"	A	H	A	B	"	"			
		"	71	H	A	"	L	B	B	B	"	"	"	"	A	L	B	B	"	"			
		"	72	H	"	B	H	B	"	A	"	"	"	"	B	L	A	A	"	"			
		"	73	H	"	"	H	A	"	A	"	"	"	"	B	L	B	A	"	"			
		"	74	L	"	"	L	A	"	A	"	"	"	"	A	H	B	B	"	"			
		"	75	H	"	"	H	B	A	B	"	"	"	"	B	L	A	B	"	"			
		"	76	L	"	"	L	B	"	A	"	"	"	"	A	H	B	A	"	"			
		"	77	"	"	"	L	A	"	B	"	"	"	"	A	H	A	A	"	"			
		"	78	"	"	"	H	A	"	A	"	"	"	"	B	L	B	B	"	"			
		"	79	"	"	A	L	B	B	B	"	"	"	"	"	H	A	B	"	"			
		"	80	"	"	"	H	B	"	A	"	"	"	"	"	L	B	A	"	"			
		"	81	"	"	"	H	A	"	B	"	"	"	"	"	"	A	A	"	"			
		"	82	H	"	"	H	A	"	A	"	"	"	"	A	H	B	B	"	"			
		"	83	L	"	"	H	B	A	B	"	"	"	"	A	L	A	B	"	"			
		"	84	H	"	"	L	B	"	A	"	"	"	"	A	H	B	B	"	"			
		"	85	H	"	"	L	A	"	B	"	"	"	"	B	H	A	B	"	"			
		"	86	H	"	"	H	A	"	A	"	"	"	"	B	L	B	B	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = 125°C and T_C = -55°C.																						
9 T_C = +25°C	t_{PLH1}	3003	87		GND	GND	OUT	2.7 V	GND	IN	GND								5.0 V	C_O to Σ_1	5	30	ns
	t_{PLH2}	Fig. 5	88	OUT	"	2.7 V		"	"	"	"					OUT	2.7 V	GND	"	C_O to Σ_2	"	"	"
	t_{PLH3}	"	89		"	"		"	"	"	"						2.7 V	GND	"	C_O to Σ_3	"	"	"
	t_{PLH4}	"	90		"	"		"	"	"	"		OUT	GND	2.7 V		2.7 V	GND	"	C_O to Σ_4	"	"	"
	t_{PLH5}	"	91		"	"		"	"	"	"	OUT		GND	2.7 V		2.7 V	GND	"	C_O to C_4	"	25	"
	t_{PLH6}	"	92	OUT	"	IN		GND	"	GND	"								"	A_2 to Σ_2	"	35	"
	t_{PLH7}	"	93	OUT	IN	GND		GND	"	GND	"								"	B_2 to Σ_2	"	"	"
	t_{PLH8}	"	94								"		OUT	GND	IN		GND	GND	"	A_4 to Σ_4	"	"	"
	t_{PLH9}	"	95								"		OUT	IN	GND		GND	GND	"	B_4 to Σ_4	"	"	"
	t_{PHL1}	"	96		GND	GND	OUT	2.7 V	GND	IN	"								"	C_O to Σ_1	"	"	"
	t_{PHL2}	"	97	OUT	"	2.7 V		"	"	"	"					OUT	2.7 V	GND	"	C_O to Σ_2	"	"	"
	t_{PHL3}	"	98		"	"		"	"	"	"						2.7 V	GND	"	C_O to Σ_3	"	"	"
	t_{PHL4}	"	99		"	"		"	"	"	"		OUT	GND	2.7 V		2.7 V	GND	"	C_O to Σ_4	"	"	"
	t_{PHL5}	"	100		"	"		"	"	"	"	OUT		GND	2.7 V		2.7 V	GND	"	C_O to C_4	"	28	"
	t_{PHL6}	"	101	OUT	"	IN		GND	"	GND	"								"	A_2 to Σ_2	"	35	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Case X 1/ Cases E,F Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Min	Test limits Max	Unit
				Σ_2	B_2	A_2	Σ_1	A_1	B_1	C_0	GND	C_4	Σ_4	B_4	A_4	Σ_3	A_3	B_3	V_{CC}				
9	t_{PHL7}	3003	102	OUT	IN	GND		GND	GND	GND	GND		OUT	GND	IN		GND	GND	5.0 V	B_2 to Σ_2	5	35	ns
T_C = +25°C	t_{PHL8}	Fig. 5	103										OUT	IN	GND		GND	GND	"	A_4 to Σ_4	"	35	"
	t_{PHL9}	"	104																"	B_4 to Σ_4	"	35	"
10	t_{PLH1}	"	105		GND	GND	OUT	2.7 V	GND	IN	GND								"	C_0 to Σ_1	"	39	"
T_C = +125°C	t_{PLH2}	"	106	OUT	"	2.7 V		"	"	"	"					OUT	2.7 V	GND	"	C_0 to Σ_2	"	"	"
	t_{PLH3}	"	107		"	"		"	"	"	"						2.7 V	GND	"	C_0 to Σ_3	"	"	"
	t_{PLH4}	"	108		"	"		"	"	"	"		OUT	GND	2.7 V		2.7 V	GND	"	C_0 to Σ_4	"	32	"
	t_{PLH5}	"	109		"	"		"	"	"	"	OUT		GND	2.7 V				"	C_0 to C_4	"	40	"
	t_{PLH6}	"	110	OUT	"	IN		GND	"	GND	"								"	A_2 to Σ_2	"	"	"
	t_{PLH7}	"	111	OUT	IN	GND		GND	"	GND	"								"	B_2 to Σ_2	"	"	"
	t_{PLH8}	"	112								"		OUT	GND	IN		GND	GND	"	A_4 to Σ_4	"	"	"
	t_{PLH9}	"	113								"		OUT	IN	GND		GND	GND	"	B_4 to Σ_4	"	"	"
	t_{PHL1}	"	114		GND	GND	OUT	2.7 V	GND	IN	"								"	C_0 to Σ_1	"	"	"
	t_{PHL2}	"	115	OUT	"	2.7 V		"	"	"	"					OUT	2.7 V	GND	"	C_0 to Σ_2	"	"	"
	t_{PHL3}	"	116		"	"		"	"	"	"						2.7 V	GND	"	C_0 to Σ_3	"	"	"
	t_{PHL4}	"	117		"	"		"	"	"	"		OUT	GND	2.7 V		2.7 V	GND	"	C_0 to Σ_4	"	"	"
	t_{PHL5}	"	118		"	"		"	"	"	"	OUT		GND	2.7 V				"	C_0 to C_4	"	35	"
	t_{PHL6}	"	119	OUT	"	IN		GND	"	GND	"								"	A_2 to Σ_2	"	40	"
	t_{PHL7}	"	120	OUT	IN	GND		GND	"	GND	"								"	B_2 to Σ_2	"	"	"
	t_{PHL8}	"	121								"		OUT	GND	IN		GND	GND	"	A_4 to Σ_4	"	"	"
	t_{PHL9}	"	122								"		OUT	IN	GND		GND	GND	"	B_4 to Σ_4	"	"	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ pins not referenced are NC.

2/ Test limits are as follows:

Symbol	Min/Max limits (mA) for circuit: A	B	C	D	E	F
I_{IL1}	-.32/-.80	-.24/-.72	-.33/-.71	-.32/-.80	-.24/-.72	-.30/-.76
I_{IL2}	-.16/-.40	-.12/-.36	-.15/-.38	-.16/-.40	-.12/-.36	-.15/-.38

3/ Alternate I_{OS} test conditions for I_{OS} test shall be as follows: Tests 47, 48, 49, 50, 51; pins Σ_2, Σ_1, Σ_4, Σ_3, C_4 respectively, shall be forced to 0.5 V. Tests 47 through 51, V_{CC} pin shall be 6.0 V.

4/ Test numbers 55 through 86 shall be run in sequence.

5/ A = ≥ 2.0 V; B = ≤0.7 V.

6/ Output voltages shall be either:
a. H ≥2.5 V minimum and L ≤0.4 V maximum when using a high speed checker double comparator.
b. H ≥1.5 V minimum and L ≤1.5 V maximum when using a high speed checker single comparator.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - Ground zero voltage potential.

V_{IN} - - - - - - - - - - - - - - - Voltage level at an input terminal.

I_{IN} - - - - - - - - - - - - - - - Current flowing into an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS83A
02	54LS283

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6.7 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an X as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Manufacturer					
	Circuit A	Circuit B	Circuit C	Circuit D	Circuit E	Circuit F
	Texas Instruments Inc.	Signetics Corporation	National Semiconductor Corp.	Raytheon Company	Fairchild Semiconductor	Motorola Inc.
01	X	X	X	X	X	X
02	X	X	X	X	X	X

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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